

$$1. \text{可交换: } x * y = 6 - 2x - 2y + xy = 6 - 2y - 2x - yx \\ = y * x$$

$$\text{令 } x = e: e * e = 6 - 2e - 2e + ee = e$$

$$\text{整理得: } (e-3)x = 6-2e \Rightarrow e=3.$$

$$\text{又: 可交换, } \therefore x * 3 = 3 * x = x.$$

$\therefore \langle R, * \rangle$ 为可交换独异点

$$\text{可结合: } x * y * z = (6 - 2x - 2y) * z \\ = 6 - 2(6 - 2x - 2y + xy) - 2z + (6 - 2x - 2y + xy)z \\ = -6 + 4x + 4y + 4z - 2xz - 2yz - 2xy + xyz \\ = x * (y * z)$$

$$5. \because a \cdot a = b, \therefore a \cdot b = a \cdot (a \cdot a) = (a \cdot a) \cdot a = b \cdot a.$$

$\therefore \langle \{a, b\}, \cdot \rangle$ 为半群, $\therefore$ 封闭.

$$\textcircled{1} a \cdot b = a \text{ 时, } b \cdot b = a \cdot a \cdot b = a \cdot (a \cdot b) = a \cdot a = b$$

$$\textcircled{2} a \cdot b = b \text{ 时, } b \cdot b = a \cdot a \cdot a \cdot a = a \cdot b \cdot a = a \cdot b = b$$

$$\text{综上所述: } b \cdot b = b$$

8. 设 $\langle S, *, e \rangle$ 为独异点, L 为 S 中左可逆元集合.

封闭性: $\forall l_1, l_2 \in L, \exists l_1', l_2' \in S, l_1' * l_1 = e, l_2' * l_2 = e$

$$\therefore \exists l_2' * l_1' \in S, (l_2' * l_1') * (l_1 * l_2) = l_2' * (l_1' * l_1) * l_2$$

$$= l_2' * e * l_2 = l_2' * l_2 = e$$

$$\therefore l_1 * l_2 \in L.$$

$$\text{令 } x = e: e * e = e \Rightarrow e \in L.$$

$\therefore L$ 为 S 的子独异点.

结合律: 原 $\langle S, *, e \rangle$ 满足结合律

故所有左可逆元素构成集合也满足结合律.

$$9. \langle \{0\}, +_6 \rangle, \langle \{0, 3\}, +_6 \rangle$$

$$\langle \{0, 2, 4\}, +_6 \rangle, \langle \mathbb{N}_6, +_6 \rangle$$

$$1. \text{群: } (1) (3) (5) (6)$$

$$\text{可交换群: } (1) (3) (5) (6)$$

2. ① 由表可知有封闭性.

② 由表可知 $*$ 可结合

③ a 为单位元

$$\textcircled{4} a * a = a, b * c = a, c * b = a,$$

所有元素均有逆元,

综上所述, $\langle G, * \rangle$ 是群.

$$8. \text{假设 } ab = ba. \text{ 又 } a^4 b = b a^5.$$

$$\therefore a^4 b = a b a^4 = a^2 b a^2 = \dots = a^5 b$$

根据消去律: $e = a$, 与 $a \neq e$ 矛盾.

故 $ab \neq ba$